OLIVER CARNEW

GRADUATE ELECTRICAL ENGINEER

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PROFILE

Hands-on electrical engineering graduate (University of Queensland, 2025) who builds and tests real hardware - power systems, battery and solar charge-control electronics, instrumentation and data-logging - and is happy on the tools, in the workshop and out on site. Backed by field experience with precision equipment, a strong safety record, and years of running my own practical businesses. I'm keen to make a hands-on start in Australia's fuel and energy-infrastructure sector and to be useful across whatever the work needs - field and site support, electrical, instrumentation, or fuel-management systems. Reliable, quick to learn, and eager to take on practical challenges.

EDUCATION

Bachelor of Engineering (Honours) - Electrical & Biomedical Engineering **2025**

The University of Queensland

- Focus areas: electrical power, control systems, embedded systems, instrumentation and signal processing.
- GPA 5.8/7. Honours project designed automated Python test pipelines measuring how search-ranking systems fail on real-world messy queries (~45% average accuracy drop across two large datasets), delivered as documented, reproducible tools.

TECHNICAL SKILLS

Electrical & Hardware: Power electronics (battery systems, solar MPPT charge control, DC-DC and USB-PD supplies); schematic capture and PCB design (KiCad, Altium Designer); hardware build, bring-up, bench test and fault-finding (oscilloscope, multimeter, logic analyser); soldering and assembly.

Instrumentation & Data: Sensor integration and data acquisition; serial interfaces (UART, SPI, I2C); embedded firmware (C/C++, FreeRTOS); monitoring and logging systems.

Software & Analysis: Python (data processing, automation), MATLAB/Simulink, SQL, JavaScript; version control (Git).

Practical & Field: Precision survey/field equipment; PC and network build and repair; clear technical documentation and user guides; workplace and electrical-safety practice.

TECHNICAL PROJECTS

Builds taken from concept through schematic, PCB and bench testing. Full portfolio available at olicar-del.github.io.

Solar Power & Battery System (MPPT): 314 Ah battery pack with safety and protection hardware, plus a solar MPPT charge controller with switched, protected load output and low-voltage disconnect - designed schematic-to-PCB.

Industrial Data Logger (DAQ): Embedded data-acquisition and logging unit in C/C++, reading multiple sensor channels over UART/SPI/I2C - optically isolated (TOSLINK) multi-board design with UI and data passthrough to PC.

USB-PD Bench Power Supply: Programmable bench supply designed schematic-to-bring-up, including power-stage design and USB Power Delivery / PPS negotiation.

ECG Analogue Front-End: Low-noise instrumentation and signal-conditioning chain, capturing live signals - precision analogue build with strict applied part Floating Body requirements.

Embedded Rhythm Game: Real-time microcontroller firmware handling live user input, file saving, audio-visual output and sub-millisecond precision.

EXPERIENCE

Undergraduate Engineer (Intern) | *Alluvium Consulting, Brisbane* **Jan – March 2025**

- Rebuilt an open-source toolkit into a documented, repeatable in-house workflow, automating data retrieval and processing to produce recurring deliverables
- Wrote clear documentation and simple tools so engineers and non-programmers could run the system themselves, reducing key-person risk.
- Quality-checked outputs against standards, presented progress weekly, and worked day-to-day within a multidisciplinary engineering team.

Survey Assistant | *AJS Surveys, Brisbane* **2021 – 2023**

- Operated and maintained precision survey equipment on residential and commercial sites alongside licensed surveyors.

- Worked to procedures across varied site and weather conditions, prioritising measurement accuracy and data integrity.
- Held responsibility for the safety of contractors and the public around live site operations.

Owner / Operator | *OC Computer Solutions* **2017-2019**

- Built, tested and repaired custom PC and network systems to customer requirements; handled diagnostics, callouts and repairs.
- Ran the full operation end to end - quoting, inventory, invoicing and customer relations.

ADDITIONAL EXPERIENCE

Owner / Disability Support Worker - OC Support Services (2023–Present): Built an independent NDIS practice to full-time operation; manage compliance, scheduling, invoicing and a support team, working ethically alongside allied-health professionals.

Private Tutor - OC Tutoring (2014–2023): Maths, Physics, Engineering and English (Years 7–12) plus piano; managed bookings, clients and invoicing.

Crew Supervisor - McDonald's Australia (2014–2023): Led teams of 2–6, trained staff and ran shifts to strict safety, quality and food-hygiene standards.

SAFETY, LICENCES & CERTIFICATIONS

Safety & Site: White Card (WHS Construction); First Aid & CPR; Manual Handling accredited; electrical-safety practice; Blue Card (WWCC); Yellow Card (NDIS).

Technical & Professional Development: MCC00072 PLC Course (Tafe QLD); Scrum Fundamentals Certified; Engineers Australia Virtual Work Experience (2023); Cert II Active Volunteering; CS50 Computer Science (Harvard); SQL Developer Certification; IBM Text Analytics 101.

Music: AMusA - Associate Diploma in Piano (AMEB, 2016).

MEMBERSHIPS & INTERESTS

Memberships: Engineers Australia (Member); UQ EUS, Electrically Based Engineering Student Society (EBESS)

Interests: Team climbing, triathlon, 4WD and camping, piano.